Jeffrey Dai

Montreal, QC | (343) 558-5988 | jeffrey.dai@mail.mcgill.ca
github.com/Jeffrey-dai17 | jeffrey-dai17.github.io/website/

EDUCATION

McGill University | Montreal, QC

Bachelor of Engineering, Computer Engineering | Expected April 2029 | **CGPA: 3.89/4.00**

Coursework: Software Systems; Software Development; Digital Logic (in progress); Algorithms and Data Structures & Model-Based Programming (Winter 2027).

WORK EXPERIENCE

IT Developer | **Canada Revenue Agency** | Ottawa / Montreal

External Fraud Management | Intern: May-August 2026; Part-time contract: September 2026-Present

- Develop and maintain Java/XML mappings that transfer information entered on application screens into a PostgreSQL database for enterprise fraud-management systems.
- Implement new mappings as requirements arrive; inspect stored data in pgAdmin and review code to identify and correct mapping errors.
- Update onboarding documentation for current software versions, clarifying outdated procedures with step-by-step instructions and screenshots.

PROJECTS

Chinatown Online | React, Node.js, Express, Socket.IO | October 2026 | GitHub

- Independently built and deployed a server-authoritative multiplayer board game for 3-5 players, with room-code joining and token-based reconnection to preserve player seats during connection drops.
- Validate trades and asset ownership on the server, expire invalidated offers after exchanges, and send each player a filtered view that keeps cash, building selections, and trade details private.
- Added unit tests for game rules, scoring, and asset transfers, plus a Playwright smoke test driving three browser sessions through room creation, trading, placement, income, and round progression.

Dishly Recipe Match | React, JavaScript | CUhacking, team of 4 | GitHub | Live demo

- Owned frontend development for a recipe-discovery app, building a responsive interface for natural-language cravings, dietary and nutrition filters, swipeable recipe cards, and recipe details.
- Connected the interface to backend APIs, preserved recipe-deck progress with browser session storage, and displayed error messages for failed requests and invalid responses.

Claw Machine | Arduino | Two-person project | 2024

- Co-built a claw machine with a 3D-printed gantry; programmed two Arduinos to drive three stepper motors for gantry movement and claw control.

LEADERSHIP & ACTIVITIES

Experience Team Lead | **McHacks** | 2025-Present

- Lead a five-person experience team for McHacks, which hosted 700 participants selected from 1,500+ applications in 2025.
- Organized an internship workshop with Compete_McGill and a networking event with Google Developer Group McGill, each drawing approximately 50 attendees.

TECHNICAL SKILLS & LANGUAGES

Programming: JavaScript, Java, Python, C, VHDL
Frameworks & Testing: React, Node.js, Express, Socket.IO, Playwright, Vitest
Tools: Git, Azure DevOps, PostgreSQL (pgAdmin), Arduino
Languages: English (fluent), French (advanced)